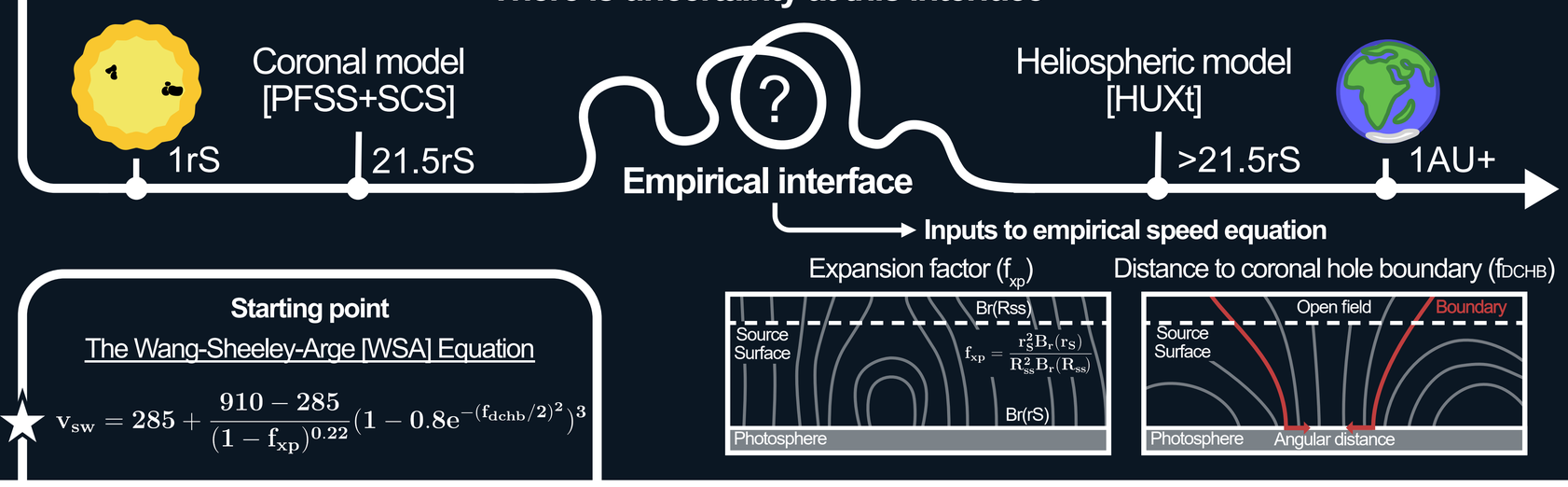
Department of Meteorology

Natural Environment Research Council | Met Office | University of Reading

Adapted empirical modelling of near-Sun solar-wind speeds for use in ensemble forecasts

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Coupled coronal-heliospheric modelling



Starting point

The Wang-Sheeley-Arge [WSA] Equation

$$v_{sw} = 285 + \frac{910 - 285}{(1 - f_{xp})^{0.22}}\left(1 - 0.8e^{-(f_{dchb}/2)^2}\right)^3$$

- **Simulating the solar wind using a coupled coronal and heliospheric model is very efficient.**
- In a coupled framework, empirical interfaces are common (e.g. WSA).
- Many uncertainties are introduced at the empirical interface making it well-suited for perturbed-boundary ensemble methods.
- **Existing equations and numbers of free parameters make perturbed-boundary methods difficult to use.**

Optimising solar-wind speed equations

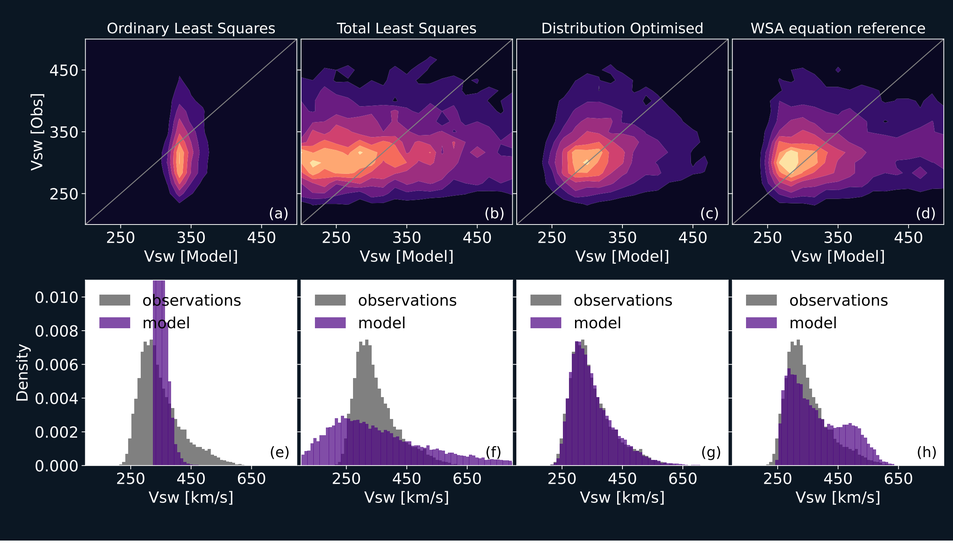
The process

- Take f_{xp} and f_{DCHB} fields from PFSS+SCS model.
- Construct speed equation candidates.
- Run multi-linear regression against training data of reconstructed OMNI observations at 21.5rS.
- Rank optimised equations and select best performing.

Result

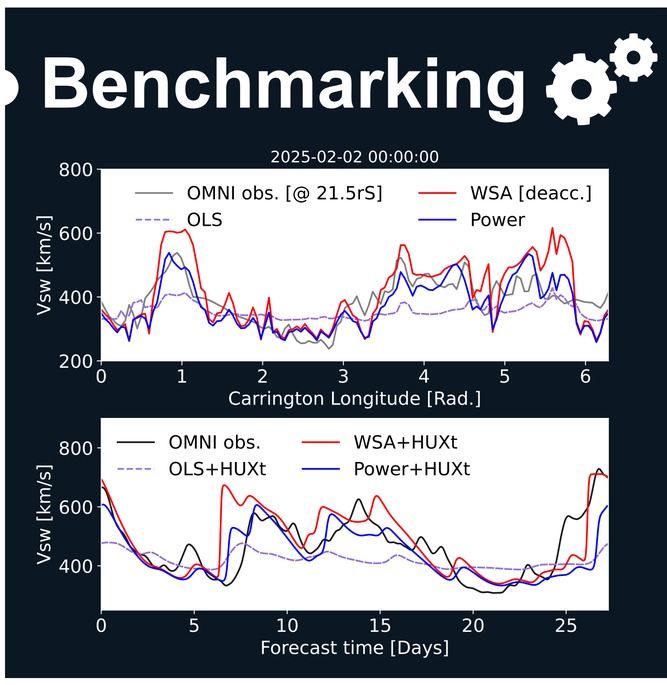
The Power equation

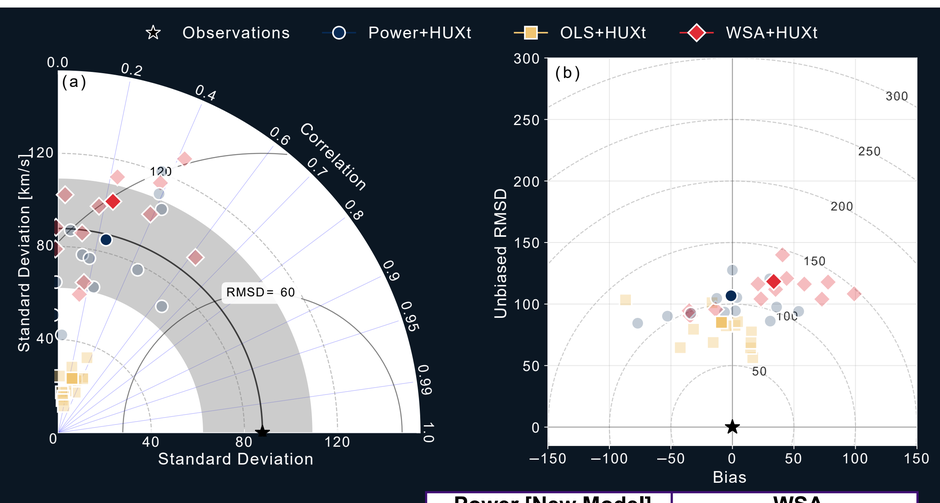
$$v_{sw} = 147f_{xp}^{-0.4} + 67f_{dchb}^{0.97} + 143$$



- **Optimisation was performed using 11-years of solar-wind speed observations, reconstructed at 21.5rS (see Owens et al 2026).**
- We explored three linear-regression techniques to highlight challenges of fitting to noisy data with large class imbalances.
- Using k-fold cross validation we produce an equation that successfully reproduced the *distribution* of observed solar-wind speeds.

Benchmarking





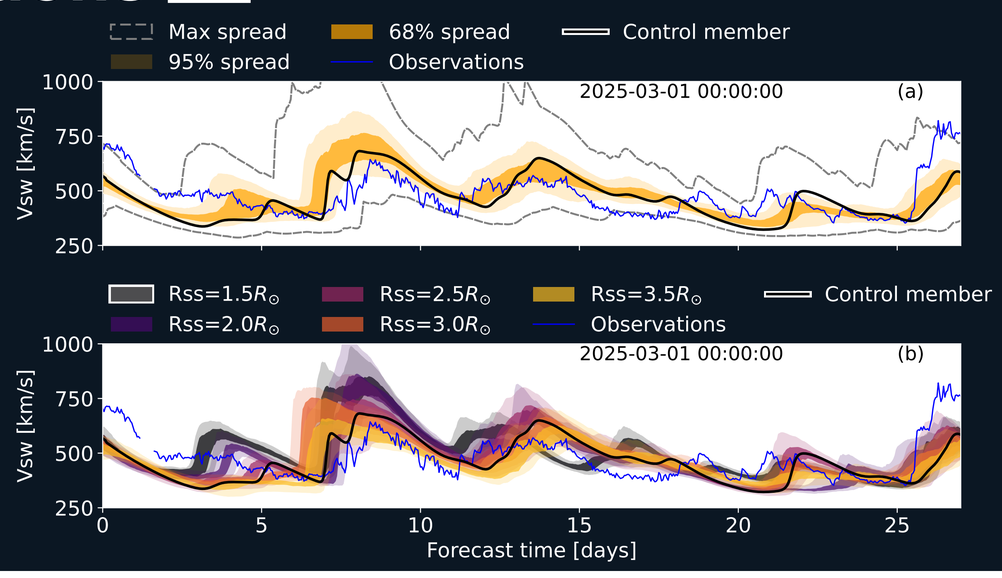
	Power [New Model]	WSA
RMSE [km/s]	106.73	122.97
Bias [km/s]	-1.11	+33.52
Correlation	0.24	0.23
Bayesian Information Criterion	154158	158861
Dynamic Time Warping	5082	5379

- **The new model performs comparably to WSA when coupled with the Heliospheric Upwind eXtrapolation with Time-dependence (HUXt) model.**
- The new equation captures observed variability closely with low levels of bias.
- The new equation is not intended to replace WSA; the internal complexity remains a key element for WSA.

Ensemble applications

Perturbed-boundary methods

- Generate 5 realisations of PFSS+SCS solutions using different source surface heights (Rss).
- Perturb speed equation to generate 10 ensemble members from each realisation.
- Apply spatial perturbations to sub-Earth path used to extract boundary conditions.
- Run boundaries through HUXt resulting in 500-member ambient solar-wind ensemble.



- **The simpler speed equation is easy to integrate into ensemble frameworks.**
- It is easy to trace the scales of perturbation directly to the ensemble spread (this is useful for calibration)
- We can combine perturbations of empirical formula with other perturbed boundary methods to sample *different* uncertainty sources.
- **Lots of potential avenues for exploring uncertainty!**

References
- Owens, M. J., et al. (2026), Driving dynamical inner-heliosphere models with in situ solar wind observations.
- Reiss, M. A., et al. (2020), Forecasting the Ambient Solar Wind with Numerical Models. II. An Adaptive Prediction System for Specifying Solar Wind Speed near the Sun.
- McGregor, S. L., et al. (2011), The distribution of solar wind speeds during solar minimum: Calibration for numerical solar wind modeling constraints on the source of the slow solar wind.

Preprint coming soon...